

Experiment - 12

Query with VIEW and INDEX

```
mysql> CREATE TABLE Department (  
->   DeptID INT PRIMARY KEY,  
->   DeptName VARCHAR(50)  
-> );  
Query OK, 0 rows affected (0.04 sec)  
  
mysql> INSERT INTO Department VALUES  
-> (1,'CSE'),  
-> (2,'IT'),  
-> (3,'ECE'),  
-> (4,'AIDS');  
Query OK, 4 rows affected (0.06 sec)  
Records: 4 Duplicates: 0 Warnings: 0  
  
mysql> CREATE TABLE Student (  
->   StudentID INT PRIMARY KEY,  
->   StudentName VARCHAR(50),  
->   DeptID INT,  
->   Age INT,  
->   Marks INT,  
->   FOREIGN KEY (DeptID) REFERENCES Department(DeptID)  
-> );  
Query OK, 0 rows affected (0.11 sec)  
  
mysql> INSERT INTO Student VALUES  
-> (101,'Rahul',1,20,85),  
-> (102,'Priya',2,21,90),  
-> (103,'Kiran',3,20,78),  
-> (104,'Sneha',1,22,88),  
-> (105,'Anitha',4,21,95);  
Query OK, 5 rows affected (0.05 sec)  
Records: 5 Duplicates: 0 Warnings: 0  
  
mysql> SELECT * FROM Student;  
+-----+-----+-----+-----+-----+  
| StudentID | StudentName | DeptID | Age | Marks |  
+-----+-----+-----+-----+-----+  
| 101 | Rahul | 1 | 20 | 85 |  
| 102 | Priya | 2 | 21 | 90 |  
| 103 | Kiran | 3 | 20 | 78 |  
| 104 | Sneha | 1 | 22 | 88 |  
| 105 | Anitha | 4 | 21 | 95 |  
+-----+-----+-----+-----+-----+  
5 rows in set (0.00 sec)
```

```
mysql> CREATE VIEW Student_View AS
-> SELECT StudentID, StudentName, Marks
-> FROM Student;
Query OK, 0 rows affected (0.05 sec)

mysql> SELECT * FROM Student_View;
+-----+-----+-----+
| StudentID | StudentName | Marks |
+-----+-----+-----+
| 101 | Rahul | 85 |
| 102 | Priya | 90 |
| 103 | Kiran | 78 |
| 104 | Sneha | 88 |
| 105 | Anitha | 95 |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> CREATE OR REPLACE VIEW Student_View AS
-> SELECT StudentID, StudentName, Age, Marks
-> FROM Student;
Query OK, 0 rows affected (0.06 sec)

mysql> SELECT * FROM Student_View;
+-----+-----+-----+-----+
| StudentID | StudentName | Age | Marks |
+-----+-----+-----+-----+
| 101 | Rahul | 20 | 85 |
| 102 | Priya | 21 | 90 |
| 103 | Kiran | 20 | 78 |
| 104 | Sneha | 22 | 88 |
| 105 | Anitha | 21 | 95 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> DROP VIEW Student_View;
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE INDEX idx_studentname
-> ON Student(StudentName);
Query OK, 0 rows affected (0.10 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> CREATE INDEX idx_studentname
-> ON Student(StudentName);
Query OK, 0 rows affected (0.10 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SHOW INDEX FROM Student;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Table | Non_unique | Key_name | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| student | 0 | PRIMARY | 1 | StudentID | A | 5 | NULL | NULL | NULL | BTREE | | | YES | NULL |
| student | 1 | DeptID | 1 | DeptID | A | 4 | NULL | NULL | YES | BTREE | | | YES | NULL |
| student | 1 | idx_studentname | 1 | StudentName | A | 5 | NULL | NULL | YES | BTREE | | | YES | NULL |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.06 sec)

mysql> SELECT *
-> FROM Student
-> WHERE StudentName = 'Rahul';
+-----+-----+-----+-----+
| StudentID | StudentName | DeptID | Age | Marks |
+-----+-----+-----+-----+
| 101 | Rahul | 1 | 20 | 85 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> DROP INDEX idx_studentname
-> ON Student;
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0
```